How-To Library

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How to Configure CyberArk Integration with Data Privacy Management

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Abstract

Informatica® Data Privacy Management can retrieve database passwords on demand from CyberArk. This article describes how you can configure Data Privacy Management to retrieve passwords from CyberArk for database data stores.

Supported Versions

- Data Privacy Management 10.4.1

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Overview

CyberArk Enterprise Password Vault secures and controls access to privileged account passwords for systems throughout an enterprise. You can configure Data Privacy Management to retrieve database passwords on demand from CyberArk. Data Privacy Management can retrieve passwords for IBM DB2 for LUW, Microsoft SQL Server, and Oracle data stores.

When you use CyberArk to manage passwords, you specify the CyberArk safe name in the data store. When you test the data store connection or you scan the data store, Data Privacy Management retrieves the password from CyberArk. Data Privacy Management does not store the password in the Data Privacy Management repository or logs.

Data Privacy Management communicates with the CyberArk vault through the CyberArk Application Identity Manager (AIM). You install the AIM on the machine that hosts Data Privacy Management. Data Privacy Management uses the CyberArk Java API to communicate with the AIM. Then, the AIM securely communicates with the CyberArk vault.

CyberArk Component Integration

Data Privacy Management integrates with the following CyberArk components:

Vault

A CyberArk vault secures and controls access to passwords. A vault contains one or more safes. When you install the CyberArk AIM, you provide details to connect and log in to the vault.

Safe

A CyberArk safe stores credentials for accounts. Data Privacy Management can retrieve passwords from multiple safes. When you configure a data store in Data Privacy Management, you specify the safe that contains the password.

Account

A CyberArk account contains the connection details and user credentials to a data source. An account is equivalent to a data store in Data Privacy Management. Each data store in Data Privacy Management must have a unique account.

Application

A CyberArk application defines which machines or users can query the CyberArk vault. The CyberArk administrator adds an application that identifies Data Privacy Management.

For more information, see the CyberArk documentation.

Data Privacy Management and CyberArk Integration Process

When you test a data store connection or you scan a data store, Data Privacy Management retrieves the data store password from the CyberArk vault. If Data Privacy Management cannot retrieve the password, the test connection or database scan job fails.

When Data Privacy Management retrieves a password from the CyberArk vault, the following high-level process occurs:

1. Data Privacy Management uses the CyberArk Java API to send a query to the CyberArk AIM.
   The query includes logic to retrieve the password for the Data Privacy Management data store.
2. The CyberArk AIM logs in to the vault with the credentials configured during the CyberArk AIM installation.
   When you install the CyberArk AIM, you provide connection details to the vault. The vault authenticates the user during the installation.
3. The vault authenticates the query.
   The authentication that the vault performs depends on the configuration in CyberArk.

   The vault verifies the following minimum requirements:

   a. The vault verifies that the application is authorized to send the query.
      After you install the CyberArk AIM, the CyberArk administrator adds an application in CyberArk. The application includes an ID that is specific to Data Privacy Management. If the application does not include the Data Privacy Management ID, the query cannot retrieve the password.

   b. The vault verifies that the application and credential providers are members of the safe that contains the account password.
      When you install the CyberArk AIM, the installer creates a unique credential provider member in the CyberArk vault. The credential provider identifies the machine that hosts the CyberArk AIM. The CyberArk administrator adds the credential provider and the application as members to the safe that includes the data store password.
4. The vault runs the query to find the password for the data store.
   Each data store in Data Privacy Management has a unique account in CyberArk. To retrieve a password for a data store, the query must find the unique account for the data store. If the query finds an account for a data store, the query returns the password.

   To determine a matching account for a data store, the query compares data store connection properties with CyberArk account properties. The query matches the data store properties with CyberArk safe, user name, address, port, and database properties. If there is a unique match for all properties, the query retrieves the password from the account.

   If multiple accounts match all properties, the query matches the account and data store names. If the query does not find a match, the query does not return a password.
5. The vault returns the password to Data Privacy Management.

Account and Data Store Mapping

A CyberArk account is equivalent to a Data Privacy Management data store. To retrieve a password for a data store, the Data Privacy Management query must find a unique account for the data store.

The following table describes the mapping of CyberArk account properties and Data Privacy Management data store properties that the query uses to determine a match:

CyberArk Account Property	Data Privacy Management Data Store Property	Description
Safe	CyberArk Safe	Required to find a unique match. Property values must be the same. The match is not case sensitive.
Username	User Name	Required to find a unique match. Property values must be the same.
Address	Host Name	Required to find a unique match. Property values do not have to be the same as you can specify an IP address, a fully qualified host name, or a host name. If the CyberArk Address and the Data Privacy Management Host Name property values are different, the query attempts to resolve the values. The query tries to match on the combination of fully qualified host name, the host name, or IP address of the host. For example, the CyberArk Address property includes an IP address and the Data Privacy Management Host Name property includes a fully qualified host name. The query does not find a match because the property values are different. The query tries to match the values on the host name, without the domain name. If there is no match, the query performs a lookup on the host name to find the IP address and tries to match on the IP address.
Port	Port	Required to find a unique match. Property values must be the same.
Database	Service Name, Database Name, or SID	Required to find a unique match. Property values must be the same. If an Oracle data store contains values for both the SID and Service Name properties, the query uses the Service Name property to match the CyberArk Database property.
Object Name	Data Store Name	Required when the query returns multiple matches based on the required match fields. If the query returns multiple matches, the query uses the account object name and the data store name to find a unique match. If the query does not return a unique match, the query cannot retrieve the password.

Steps to Configure CyberArk Integration with Data Privacy Management

To configure integration between CyberArk and Data Privacy Management, install the CyberArk Application Identity Manager (AIM). In the Data Privacy Management data store, set the CyberArk safe that contains the data store password.

The CyberArk administrator adds an application in CyberArk and completes additional steps in CyberArk. This task includes information required to integrate with Data Privacy Management and is not a comprehensive list of steps that the CyberArk administrator performs.

To configure integration with CyberArk, perform the following high-level steps:

1. Install the CyberArk Application Identity Manager (AIM) on the machine that hosts Data Privacy Management. Data Privacy Management uses the CyberArk AIM to communicate with the CyberArk vault. When you install the CyberArk AIM, you configure the connection to the CyberArk vault and provide credentials to log in to the CyberArk vault.

   For more information about how to install the CyberArk AIM, see the CyberArk documentation.

2. Copy the following CyberArk .jar file:

   ```
   /opt/CARKaim/sdk/JavaPasswordSDK.jar
   ```

   to the following Data Privacy Management directories:

   ```
   /$INFA_HOME/services/shared/jars/thirdparty
   /$INFA_HOME/java/jre/lib/ext
   ```

3. Restart the Informatica domain.

4. The CyberArk administrator adds an application for Data Privacy Management. Verify the following information:

   - The Application Name for Data Privacy Management is `Informatica_SecureAtSource`. If the application name is not set to this value, Data Privacy Management cannot retrieve passwords from the CyberArk vault.
   - Data Privacy Management does not require the application to allow extended authentication restrictions. However, if the administrator enables extended authorization restrictions, the administrator must set values specific to Data Privacy Management.
   - If the application is restricted based on operating system users, add the operating system user that installed Data Privacy Management. If the application is restricted based on the IP address, host name, or DNS, add the machines that host Data Privacy Management and Hadoop.

   The CyberArk administrator adds the credential provider and the Data Privacy Management application as members of the safe that stores data store passwords.

   During the AIM installation, the installer creates a unique credential provider member in the CyberArk vault. The installer uses the following syntax to create the member name:

   ```
   Prov_<CyberArk AIM host name>
   ```

   The CyberArk administrator repeats this step for each safe that stores data store passwords.

5. Verify that the credential provider member has the following minimum authorizations:

   - Retrieve Accounts
   - List Accounts
   - View Safe Members

6. Verify that the Data Privacy Management application member has **Specify the Retrieve Accounts** authorization.

7. In Data Privacy Management, add the CyberArk safe name in the data store properties. Test the connection to verify that Data Privacy Management can retrieve the password from the CyberArk vault and connect to the data source.
   Repeat this step for each data store that has a password stored in the CyberArk vault.

Note: If the CyberArk administrator configures caching for the CyberArk AIM, verify that the refresh interval for the cache is frequent. If passwords change often in CyberArk and the refresh interval is infrequent, Data Privacy Management might not retrieve the latest password.

Frequently Asked Questions

Review the following frequently asked questions to understand how to integrate CyberArk with Data Privacy Management:

Which data stores can Data Privacy Management use to retrieve passwords in CyberArk?

Data Privacy Management can retrieve passwords from CyberArk for the following database data stores:

- IBM DB2 for LUW
- Microsoft SQL Server
- Oracle

When does Data Privacy Management retrieve a password in CyberArk?

Data Privacy Management retrieves the password for a data store if a CyberArk safe is configured in the data store properties. Data Privacy Management retrieves the password from the CyberArk vault when you test a data store connection and when you run a database scan job. The scan job retrieves the password each time the scan job connects to the source. The scan job connects to the source during the Load Catalog, Profiling, and Collect Row Count job steps.

What happens if Data Privacy Management cannot retrieve the password from CyberArk for a data store?

If Data Privacy Management cannot retrieve a password, the data store test connection and the scan job fail with a connection error.

Does Data Privacy Management store the password?

The password is never stored in the Data Privacy Management repository or logs. Data Privacy Management instantaneously uses the password to establish a JDBC connection and then discards the password. The password is temporarily stored in encrypted format in the Informatica Platform during a scan job and is removed when the scan job fails, completes, pauses, or stops.

What happens if an account is defined in multiple safes?

CyberArk returns the password only from the safe that is configured in the data store properties.

The CyberArk account is enabled for automatic password management. Can Data Privacy Management still access the password?

Yes. Data Privacy Management retrieves passwords for accounts that are enabled for automatic password management. If CyberArk is changing a password while a scan job runs, the scan job fails. When you resume the job, Data Privacy Management retrieves the latest password from CyberArk.

How I do I remove CyberArk integration for a data store?

The integration is enabled when you specify a CyberArk safe name in the data store properties. To disable the CyberArk integration for a data store, remove the CyberArk safe name from the data store properties. Add the password and save the data store.

Author

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